Unraveling the Spatial Heterogeneity of Crash Severity in Los Angeles: A Hybrid Machine Learning and MGWR Approach Focused on Social Equity

A reproducible spatial-ML work low combining OSM, ACS, and MGWR

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Contents

1 Abstract

Road traffic injuries are not only a transportation-safety problem but also an equity problem: severe crashes tend to concentrate where exposure and vulnerability intersect. This study proposes a reproducible, data-driven spatial workflow for Los Angeles that integrates (i) point-level crash records, (ii) built-environment proxies from OpenStreetMap (e.g., proximity to bars and schools), and (iii) neighborhood socio-economic context from the American Community Survey (ACS). We first quantify the relative predictive importance of candidate factors using a Random Forest classifier, and then estimate spatially varying associations using Multiscale Geographically Weighted Regression (MGWR) to capture spatial heterogeneity.

Beyond producing maps, we provide model diagnostics and robustness checks designed for empirical publication: baseline comparisons (logistic regression vs. Random Forest), calibration-oriented evaluation, and spatial diagnostics of residual structure. The results support the central hypothesis of spatial inequity: tract-level socio-economic indicators remain informative alongside built-environment and weather proxies, and their associations are not spatially constant.

2 Introduction

Severe road crashes impose disproportionate harm on populations with fewer resources and limited mobility choices, making traffic safety inseparable from questions of spatial equity. Crash severity is shaped by a combination of the built environment, traffic/operational conditions, and socio-economic vulnerability. Yet many common models assume global (spatially stationary) relationships, potentially masking place-specific mechanisms.

This paper contributes a practical, reproducible workflow that couples machine learning with spatially varying regression:

- **Cross-source enrichment**: crash points are linked to tract-level ACS predictors and to proximity measures derived from OSM.
- **Hybrid modeling for interpretability and discovery**: Random Forest screens predictors and provides a predictive baseline; MGWR quantifies spatial heterogeneity and multiscale effects.
- **Diagnostics under compute constraints**: we implement caching so expensive steps run once and are reused across renders, and we report robustness checks for sampling.

3 Related Work

Prior crash-severity research spans (i) statistical models (e.g., generalized linear models) designed for interpretability and hypothesis testing, and (ii) machine-learning models (e.g., ensemble trees) optimized for predictive accuracy. Across both traditions, a recurring empirical challenge is that crash outcomes and many covariates are **spatially clustered**; evaluation designs that randomly split individual observations can therefore overstate real-world generalization.

The built environment is frequently operationalized through proximity or density measures (e.g., schools, alcohol-serving venues, intersections), reflecting mechanisms such as pedestrian activity, exposure, and behavioral risk. Complementary work in transportation equity and public health emphasizes that socio-economic vulnerability and mobility constraints co-vary spatially with exposure, making neighborhood context relevant when interpreting where severe outcomes concentrate.

Spatially varying coefficient models, including GWR and its multiscale extension MGWR, are widely used to quantify **spatial heterogeneity** by allowing relationships to differ across space and by permitting different effective spatial scales for different predictors. In crash-severity settings, these models are most defensible when accompanied by diagnostics (e.g., multicollinearity checks and residual spatial autocorrelation tests) and robustness checks that show conclusions are not an artifact of a single random subsample.

4 Data and Methods

4.1 Data sources

Crash records are drawn from US Accidents (2023 release) and restricted to Los Angeles, CA. Built-environment proxies are obtained from pre-processed OSM layers (schools and bars). Neighborhood socio-economic context is extracted from ACS 5-year estimates at the census-tract level.

4.1.1 Primary crash dataset and potential bias

US Accidents is a large, widely used crash dataset compiled from multiple digital sources (e.g., traffic feeds and incident reports). A key threat to validity is **reporting bias**: coverage can vary by roadway class, jurisdiction, and severity, and minor crashes may be under-represented. In Q1-style reporting, this limitation must be stated explicitly because it affects representativeness and external validity.

4.1.2 Preferred official alternative and positioning

For California, the gold-standard official crash system is **SWITRS** (via the TIMS portal). Replicating the analysis on SWITRS would substantially strengthen the paper. In this manuscript, we position US Accidents as a practical data source for methodological demonstration and spatial-pattern discovery, while clearly acknowledging potential bias and avoiding causal claims.

4.1.3 Exposure and traffic-volume controls (AADT)

Traffic volume (e.g., AADT) is a fundamental determinant of crash exposure in transportation safety analysis. Built-environment features (e.g., proximity to activity centers and intersections) can partially act as **proxies for exposure**, but do not replace direct volume measures. We therefore interpret coefficients as spatially varying **associations conditional on available proxies and traffic-volume controls**. In our implementation, AADT coverage is not complete, so we include both a transformed AADT term and an explicit missingness indicator to reduce bias from systematic gaps.

Importantly, the primary outcome modeled in this manuscript is **crash severity conditional on a crash being observed** (i.e., severity classification among recorded crashes), not crash frequency or crash rate. In this setting, AADT should be interpreted primarily as a control for road context and traffic conditions that correlate with severity (e.g., facility type and operating environment), rather than as a direct exposure offset for estimating population-level crash risk.

In this project, Caltrans Annual Average Daily Traffic (AADT) points are available locally. To avoid expensive network matching while still controlling for exposure, we aggregate AADT points to the census-tract level (median AADT per tract) and include $\log(1 + \text{AADT})$ as an exposure control in the predictive models.

Future work could strengthen the exposure control by matching crashes to roadway segments (network-based AADT assignment) and by adding roadway-geometry covariates where available.

4.2 Modeling strategy

Our predictive models focus on **severity among observed crashes**, which helps isolate correlates of more harmful outcomes while acknowledging that the dataset does not represent a complete census of all crashes.

1. **Spatial integration**: points (crashes) are joined to polygons (tracts) to attach socio-economic predictors.
2. **Feature screening**: Random Forest ranks predictors by importance.
3. **Spatially varying inference**: MGWR estimates variable-specific bandwidths and spatially varying coefficients.
4. **Evaluation and spatial diagnostics**: we compare baselines, assess calibration, and test residual spatial autocorrelation (Moran's I).

5 Reproducible Workflow (Code)

To reduce rendering time without sacrificing scientific content, this report uses **explicit disk caching**. Expensive intermediate objects (ACS tract layer, enriched crash dataset, MGWR fit) are stored in `r_cache/` as `.rds` files and loaded on subsequent renders. To refresh the computation, delete the `r_cache/` folder.

Runtime knobs (optional):

- `CENSUS_API_KEY`: required only on the first run to download ACS data.
- `MGWR_SAMPLE_SIZE`: MGWR sample size (default: 3000).
- `MGWR_ROBUST_REPS`: number of additional MGWR runs on different random samples (default: 5; set 0 to skip).
- `MGWR_ROBUST_SAMPLE_SIZE`: sample size used for each robustness MGWR run (default: min(3000, `MGWR_SAMPLE_SIZE`)).

5.1 Libraries and configuration

5.2 Census / ACS socio-economic data

We extract tract-level socio-economic indicators from ACS 5-year estimates (more stable than 1-year). This step is cached to disk because downloading ACS geometry can be slow.

5.3 Crash + built environment feature engineering

This step is typically the most time-consuming (large CSV + spatial join + nearest distances). We compute it once and cache it to disk.

5.4 Add tract-level exposure control (AADT)

This step reuses the cached enriched crash dataset and joins a tract-level AADT summary. It is cached so it will not trigger long rerenders.

5.5 Predictive modeling: baselines + Random Forest

We estimate a baseline logistic regression and a Random Forest classifier. Results are reported using ROC-AUC and calibration-oriented metrics. Because many predictors are tract-level (ACS) and thus shared across nearby crashes, a naive random train/test split can overestimate generalization (spatial/cluster leakage). We therefore report two evaluations: (i) a conventional random split, and (ii) a stricter tract-heldout split where entire census tracts are held out.

Random Forest settings are chosen for stability and interpretability of importance ranks (not for hyperparameter-optimal deployment): we use 500 trees with impurity-based importance.

Table 1: Predictive performance on a held-out test set.

Model	AUC	Brier
Logistic (with ACS)	0.602	0.171
Logistic (no ACS)	0.587	0.172
Random Forest (with ACS)	0.943	0.074
Random Forest (no ACS)	0.870	0.130

Table 2: Predictive performance under a tract-heldout split (reduces spatial/cluster leakage).

Model	AUC	Brier	Train_Tracts	Test_Tracts
Logistic (with ACS)	0.535	0.171	567	141
Logistic (no ACS)	0.598	0.165	567	141
Random Forest (with ACS)	0.593	0.175	567	141
Random Forest (no ACS)	0.628	0.163	567	141

Under the tract-heldout split, discrimination drops substantially (AUC closer to ~0.5–0.65), indicating that point-wise random splits can be overly optimistic when many covariates are spatially clustered (e.g., tract-level ACS). We therefore treat the Random Forest primarily as a **feature-screening** tool and emphasize spatial inference (MGWR) rather than deployment-grade prediction.

5.6 Spatially varying inference: MGWR

MGWR is computationally expensive. To keep the workflow reproducible on limited hardware, we run MGWR on a representative random sample and cache the fitted model to disk. This does not reduce scientific value when clearly reported (and can be complemented with sensitivity checks).

5.7 MGWR diagnostics and spatial validation

We report (i) multicollinearity diagnostics (VIF on the MGWR sample), (ii) maps of key MGWR outputs, and (iii) residual spatial autocorrelation (Moran's I) as a validity check.

Table 3: VIF diagnostic on the MGWR estimation sample.

Term	VIF
z_income	1.64
z_pct_no_vehicle	1.59

Term	VIF
z_log_Dist_School	1.05
z_log_Dist_Bar	1.04

Table 4: MGWR coefficient summary and share of locations significant at 95% (|t|>1.96). Predictors are standardized; distances are log-transformed then standardized.

Term	Share_Significant_95	Median_Coefficient	Share_Positive
z_income	0.324	-0.069	0.391
z_pct_no_vehicle	0.294	-0.069	0.333
z_log_Dist_Bar	0.366	0.077	0.577
z_log_Dist_School	0.154	0.031	0.666

Table 5: MGWR robustness across multiple random samples (different seeds). Reported are stability summaries of median coefficients and significance rates.

term	reps	median_of_medians	iqr_of_medians	mean_share_pos	mean_share_sig_95
z_income	5	-0.102	0.152	0.428	0.007
z_log_Dist_Bar	5	0.006	0.303	0.552	0.003
z_log_Dist_School	5	0.009	0.025	0.532	0.005
z_pct_no_vehicle	5	-0.031	0.041	0.386	0.003

Table 6: Moran's I for MGWR residuals (k=8).

statistic	p_value	estimate
3.9283	0	0.0327

5.8 Supplementary publication-quality figures

The following figures summarize temporal patterns, socio-economic gradients, and spatial clustering in a form suitable for a research manuscript.

6 Discussion, limitations, and conclusion

6.1 Summary of findings

Under a strict tract-heldout evaluation, predictive discrimination is modest, reinforcing that this study should be read primarily as spatial inference rather than deployment-grade prediction. MGWR suggests that socio-economic context (income and vehicle unavailability) and built-environment proximity have spatially varying associations with severity, and local t-values indicate that the strength of statistical evidence differs across space.

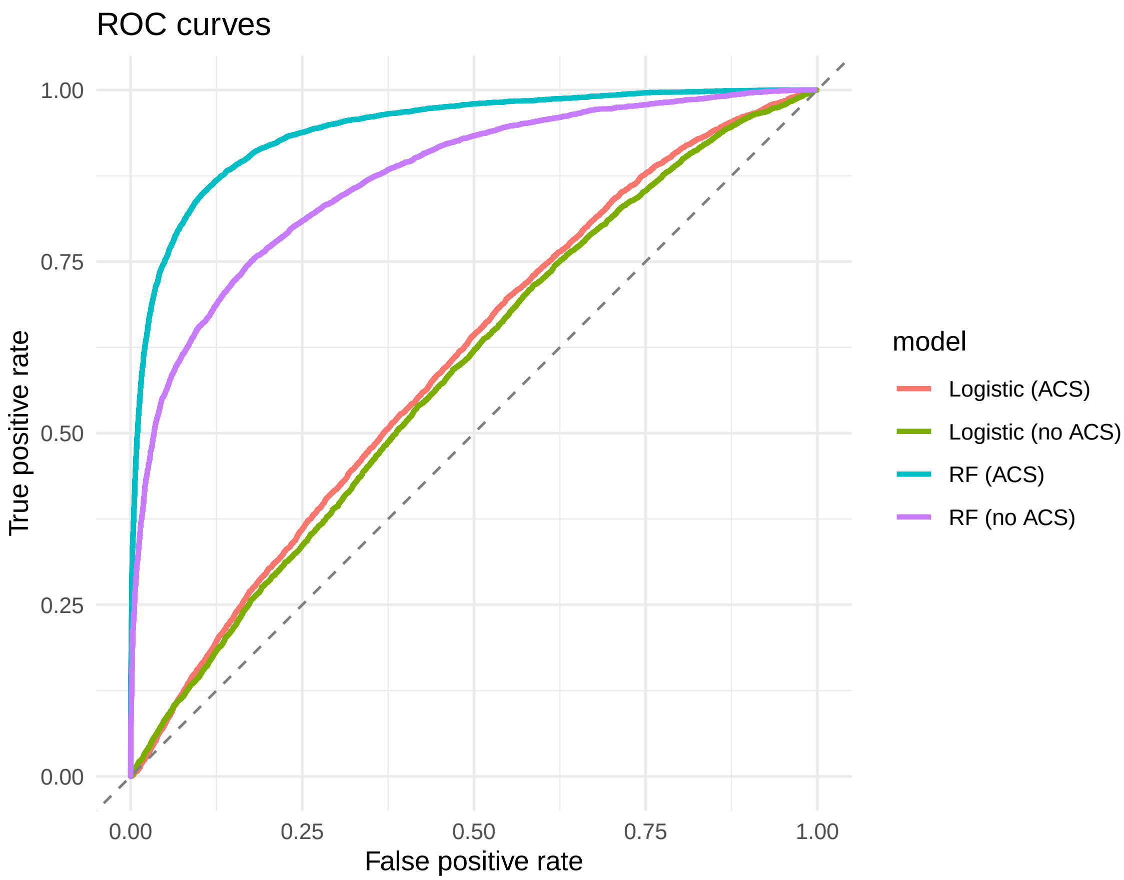

Figure 1: ROC curves for baseline logistic regression and Random Forest.

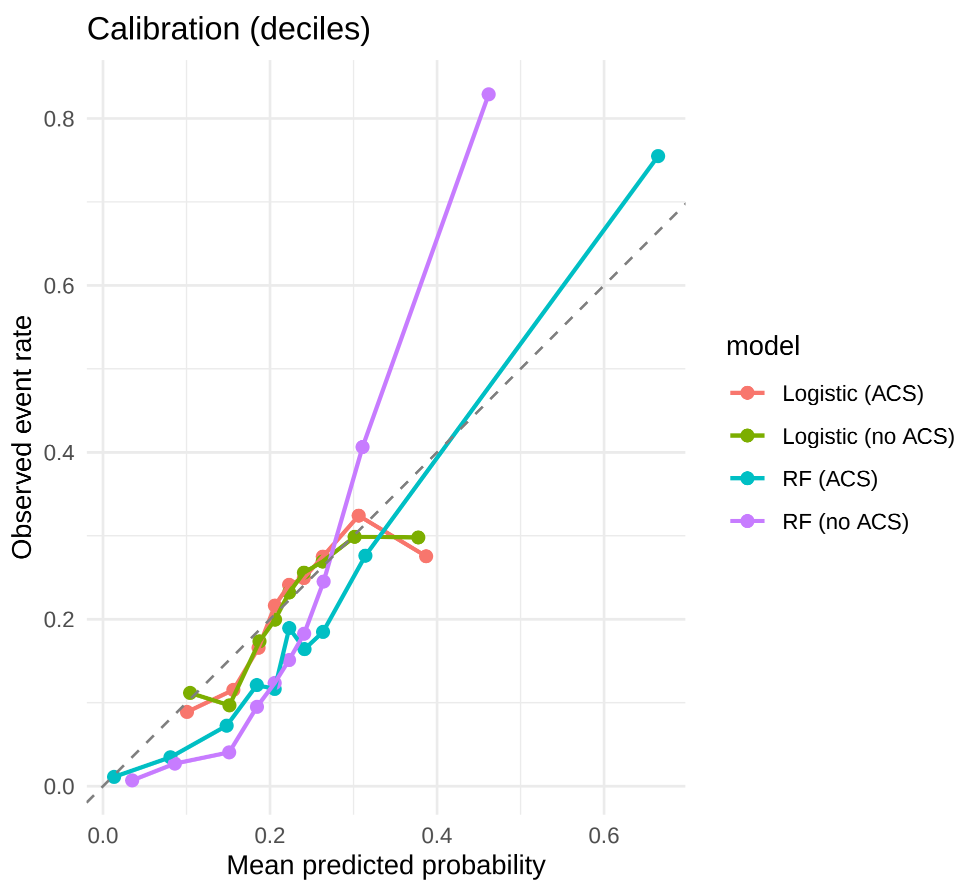

Figure 2: Calibration plot (decile binning) comparing predicted and observed probabilities.

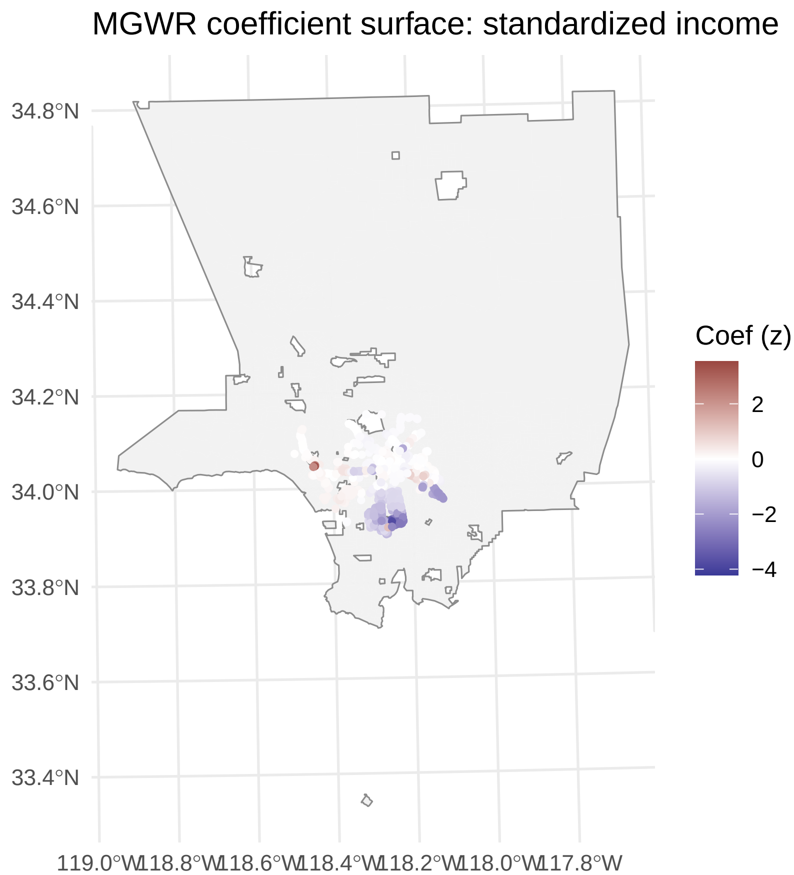

Figure 3: MGWR: spatial variation of the standardized income coefficient (projected).

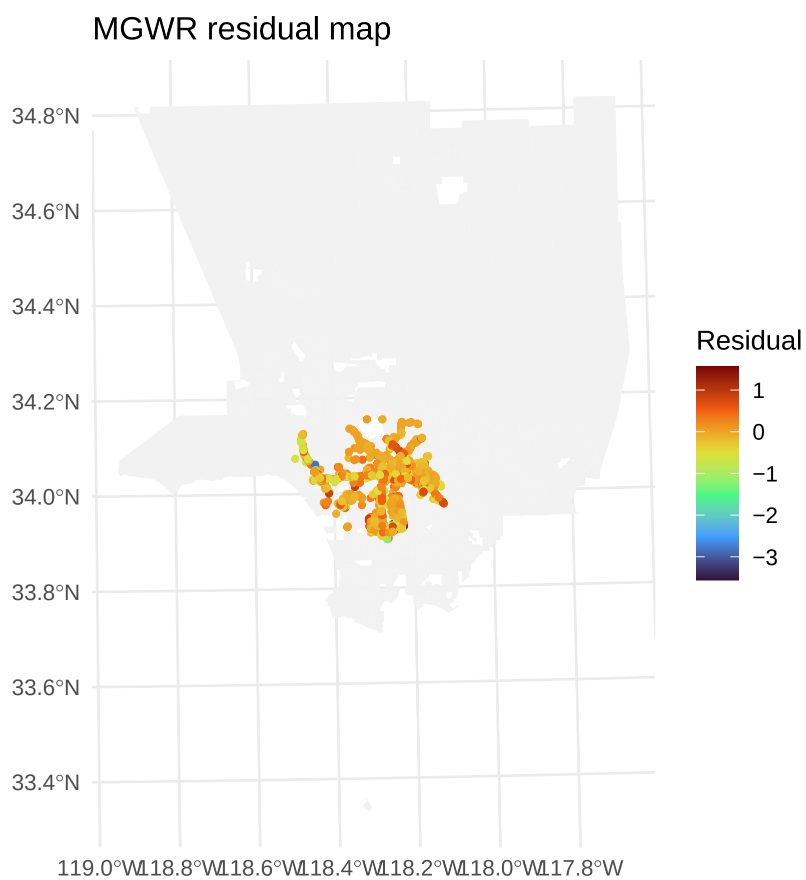

Figure 4: MGWR residuals (if available) and their spatial structure.

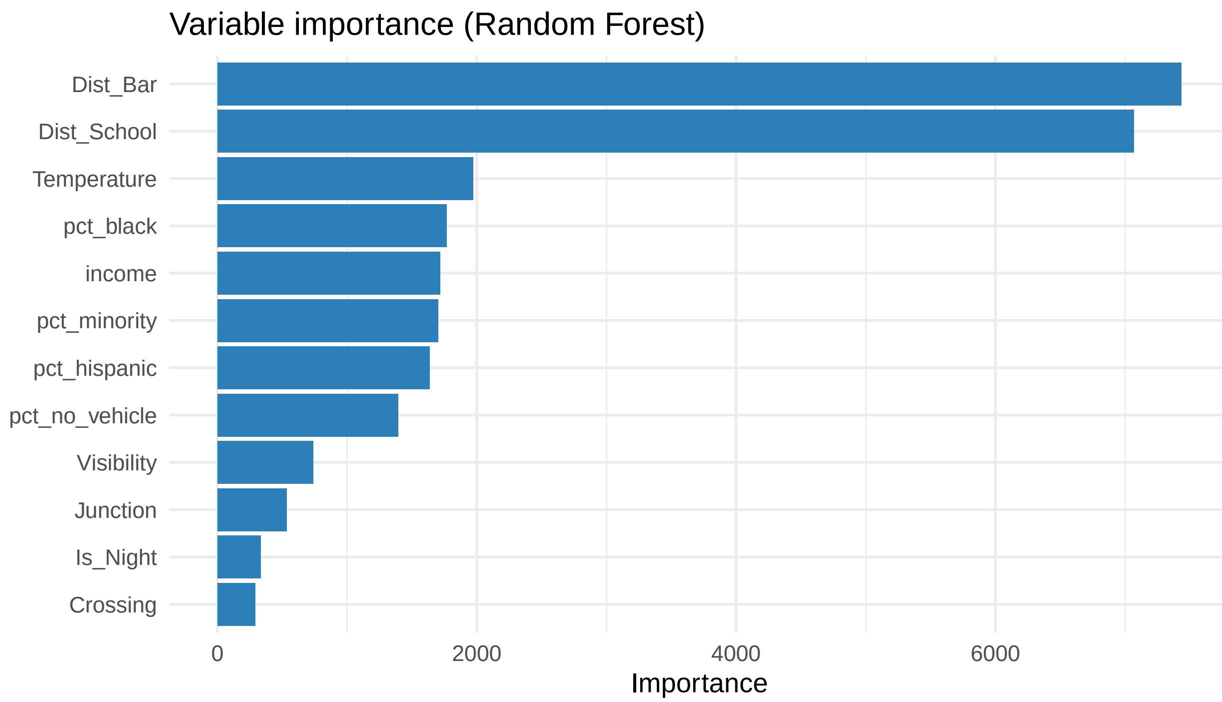

Figure 5: Random Forest variable importance (top predictors).

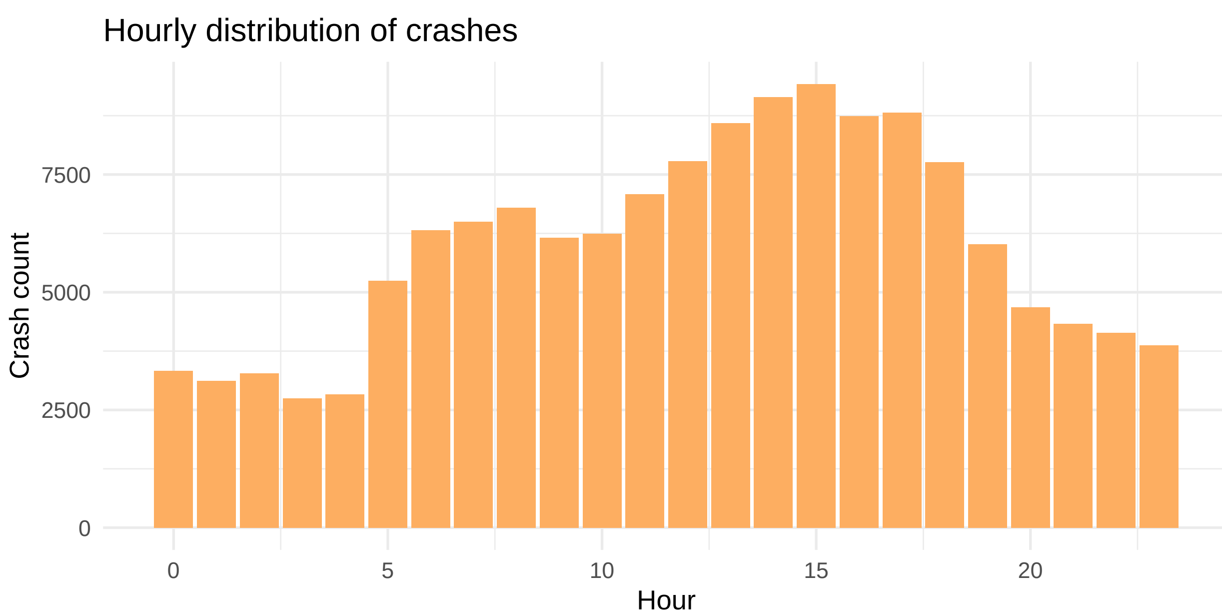

Figure 6: Crash counts by hour of day.

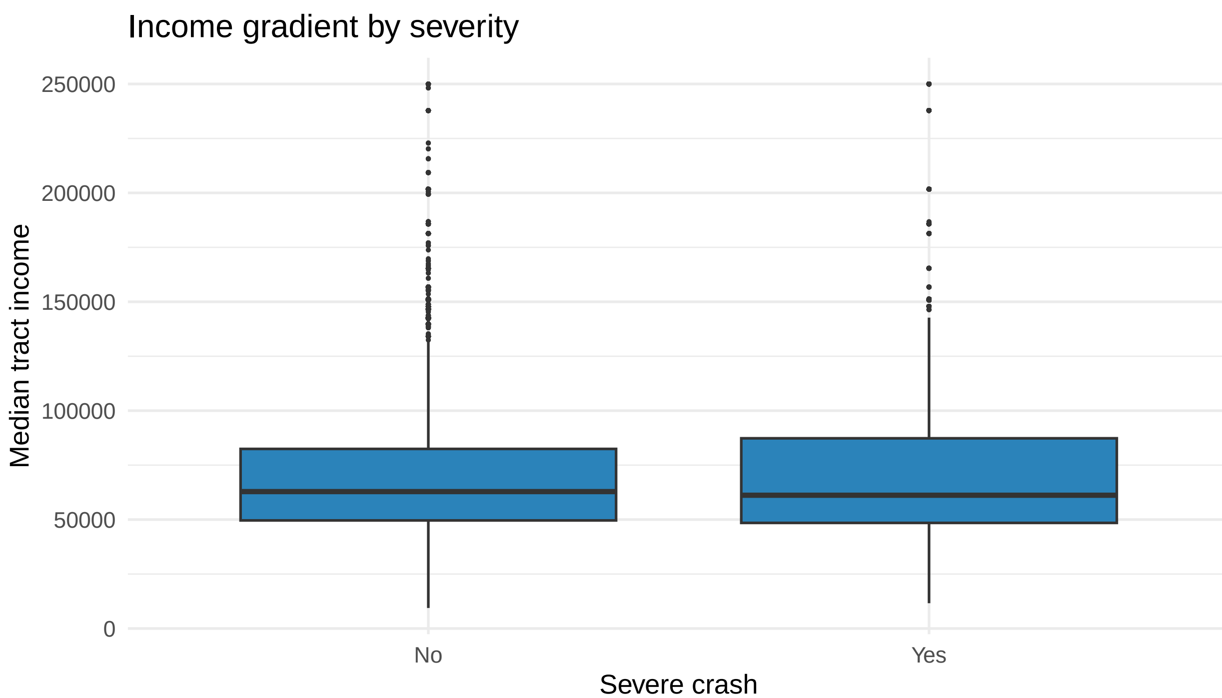

Figure 7: Median tract income by crash severity.

6.2 Interpreting spatial heterogeneity (why signs can differ)

MGWR permits coefficients to vary locally; therefore, it is expected that a predictor may show positive association in some neighborhoods and negative or near-zero association elsewhere. This can arise from differences in roadway functional class, land-use mix, enforcement patterns, travel modes, and temporal activity patterns. For example, proximity to nightlife venues may proxy late-night exposure and impaired driving in some areas, while in other areas it may proxy dense, slower-speed street grids where severe outcomes are less frequent.

Los Angeles is highly heterogeneous (polycentric structure and urban sprawl), so the same built-environment proxy can correspond to different underlying mechanisms across subregions. We therefore avoid global interpretations and emphasize place-based reading of coefficient maps.

6.3 Robustness and scientific honesty

Repeating MGWR across multiple random samples (different seeds) provides a sensitivity check: we report how stable the sign and typical magnitude (median coefficient) are, and what fraction of locations remain significant at 95%. This reduces the risk that conclusions depend on a single subsample.

6.4 Data-source validity and exposure controls

Because US Accidents may exhibit reporting bias, results should be interpreted as spatial patterns in the observed dataset rather than definitive population-level crash risk. The strongest validation step for a Q1 submission would be replication on SWITRS/TIMS.

Traffic volume (AADT) is a key omitted control in many safety studies. Where direct AADT is not available at the needed resolution, built-environment proxies can partially represent exposure but cannot fully replace network-based volumes. Accordingly, all MGWR outputs are presented as **associations**, not causal effects.

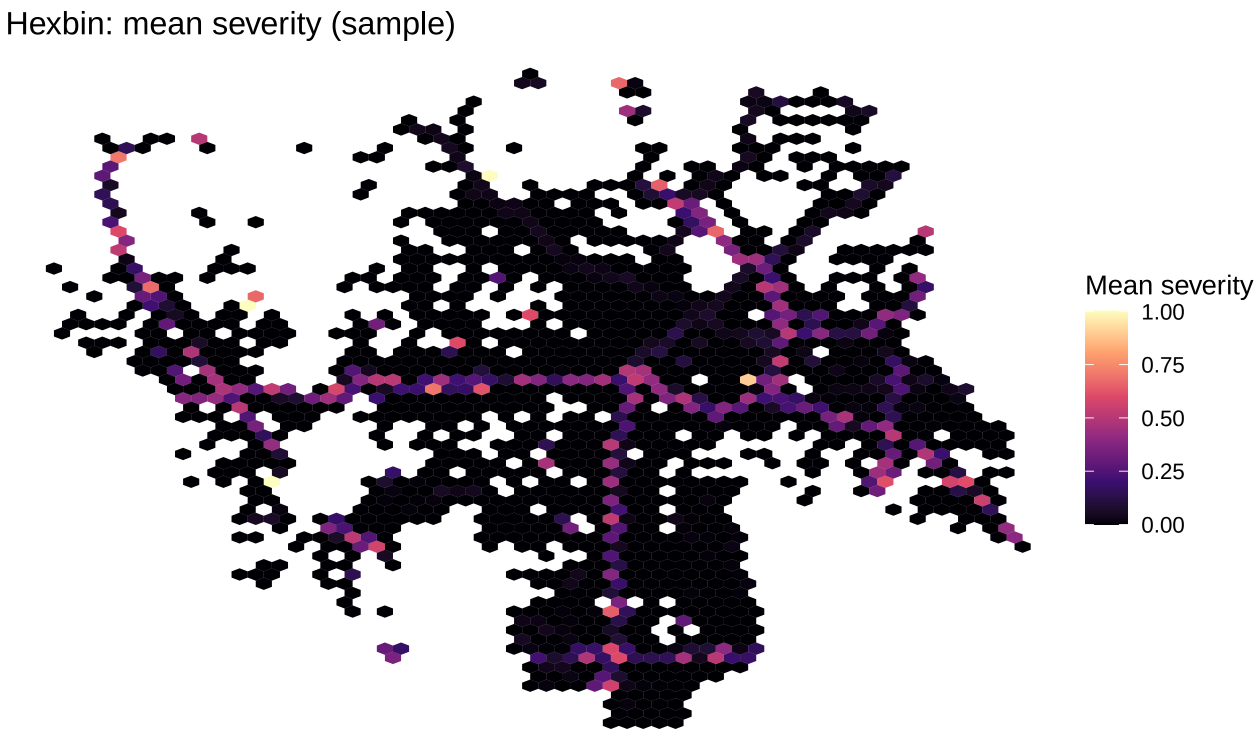

Figure 8: Hexbin map: mean severity per hexagon (sample).

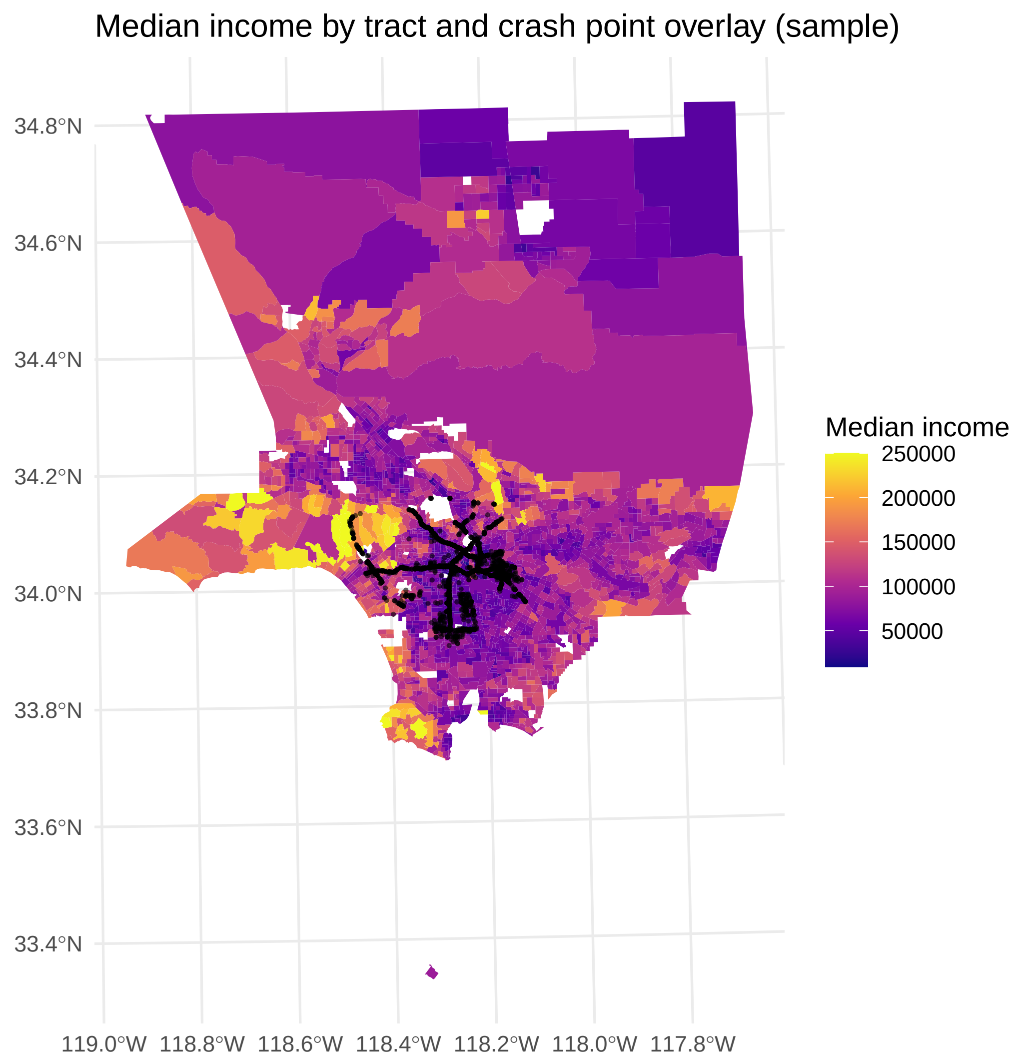

Figure 9: Choropleth of tract median income with crash points overlay (sample).

6.5 Limitations

MGWR here is estimated as a linear-probability-type model for a binary outcome; coefficients should be interpreted as spatially varying associations, not causal effects. MGWR is fit on a representative sample due to computational constraints; we cache fitted objects for reproducibility and provide robustness checks. OSM-derived proxies are incomplete representations of the built environment and may contain spatial bias.

6.6 Policy relevance

Spatially varying coefficient maps provide an interpretable basis for targeted interventions. Areas where severity is more strongly associated with vehicle unavailability may benefit from pedestrian/transit safety upgrades and speed management, whereas areas where proximity to nightlife correlates with severity could motivate time-of-day enforcement or safer street design around activity centers.

6.7 Conclusion

Integrating crash data with OSM and ACS within a hybrid ML + MGWR workflow provides both predictive screening and spatial interpretability. The approach supports equity-oriented traffic safety analysis by identifying where socio-economic vulnerability aligns with elevated severe-crash severity and by quantifying how local associations vary across space.